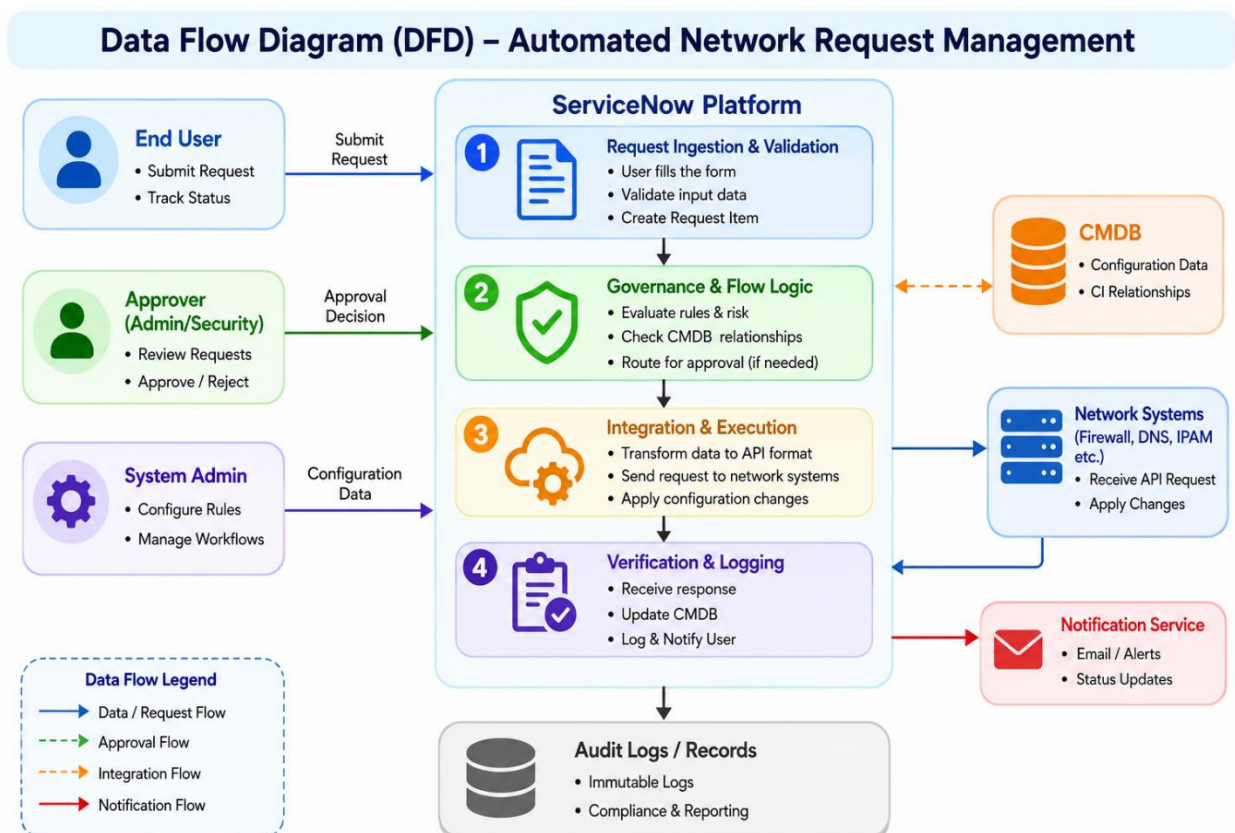


Requirement Analysis

Data Flow Diagram & User Stories

Date	29-06-2026
Project Name	Automated Network Request Management
Team ID	
Maximum Marks	

Part 1: Data Flow Diagram (DFD)



Phase 1: Request Ingestion & Data Validation

- Data Origin: End user submits a network request through a ServiceNow Service Portal form.
- Data Captured: Inputs such as Source IP, Destination IP, Port number, request type, and justification.
- Validation Process:
 - Client-side validation ensures required fields are filled.
 - Server-side UI Policies and Data Policies validate formats (IP regex, port range, mandatory fields).
- Record Creation: After validation, a request item (sc_req_item) is created in ServiceNow.

Phase 2: Governance & Workflow Decision Logic

- Process Engine: Flow Designer evaluates incoming request data.
- Rule Evaluation:
 - Checks CMDB relationships and service dependency mappings.
 - Applies risk scoring based on CI criticality and request type.
- Decision Flow:
 - High-risk requests → Routed for manual approval (Manager/Security Team).
 - Low-risk requests → Auto-approved based on predefined rules.
- Outcome: Approved request proceeds to automation pipeline.

Phase 3: Integration & Infrastructure Execution

- Integration Layer: Integration Hub converts ServiceNow data into API-ready format (JSON/XML).
- Outbound Communication:
 - REST API calls or Ansible orchestration triggers are initiated.
 - Payload sent to network systems (e.g., firewall, DNS, SDN controllers).
- Execution:
 - Target systems apply configuration changes (firewall rules, IP allocations, port access updates).

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Phase 4: Verification & Audit Logging

- **Response Capture:** Infrastructure systems return execution status (success/failure codes + logs).
- **Processing:**
 - ServiceNow parses response data.
 - Updates CMDB with latest configuration state.
- **Audit & Compliance:**
 - Immutable logs stored for tracking and auditing.
- **User Notification:**
 - End user receives confirmation email/notification about request completion.

Part 2: User Stories

Here is a set of comprehensive, production-ready Agile User Stories tailored for the **Automated Network Request Management (ANRM)** ServiceNow project, structured in the requested tabular format.

ServiceNow ANRM User Stories & Tasks

User Type	Functional Requirement	User Story / Task	Acceptance Criteria	Priority

End-User <i>(Developer / Requestor)</i>	Self-Service Request Intake	As an end user, I want to submit a network access request through a portal so that I can get required access without manual coordination.	• Form must include mandatory fields: Source IP, Destination IP, Ports, Protocol (TCP/UDP), and Business Justification. • Form must dynamically show/hide advanced routing choices based on user role.	High
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IT Ops Manager <i>(Approver)</i>	Automated Governance / Approval Routing	As an IT Operations Manager, I want low-risk requests auto-approved and high-risk requests routed to my queue, So that standard requests aren't delayed by manual approvals.	• Flow Designer must evaluate request criteria against a predefined risk matrix. • Requests with known standard ports (e.g., 443) auto-approve instantly. • High-risk ports (e.g., 22, 3389) trigger an email/system approval notification to the manager.	High
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ServiceNow Dev (Fulfiller)	Integration Hub API Integration	As a ServiceNow Developer, I want Integration Hub to trigger REST API payloads to the network controller, So that configuration changes are applied programmatically.	• Upon approval token, Integration Hub must compile form data into a valid JSON payload. • Payload must be securely sent to the target API endpoint (e.g., Palo Alto/Ansible) using encrypted credentials. • System must capture synchronous HTTP response codes (200 OK for success).	Critical
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Network Administrator <i>(Auditor)</i>	Automated Verification & CMDB Logging	As a Network Admin,I want the system to test the connection and log details to the CMDB,So that our inventory remains accurate and compliant.	• Post-execution step must verify connectivity status. • On success, the script must write a configuration relationship record back to the target CI in the CMDB. • The RITM (Requested Item) ticket must lock down an uneditable audit log containing who requested, who approved, and API logs.	Medium
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End-User <i>(Developer / Requestor)</i>	Real-Time Notification & Visibility	As an End-User, I want real-time status updates on my request, So that I don't have to send follow-up emails to see if it's done.	• System must trigger automated notifications (Email or Slack/Teams webhook) at three milestones: Submitted, Approved/In-Flight, and Fulfilled/Closed. • If an API push fails, notification must alert NetOps with the failure logs.	Low
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